

Chronology Changes Analytic Type in a Dyadic-Solenoid Skew Product

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Abstract

We give an explicit example in which chronological order changes the analytic type of a dynamical zeta factor. Let the full two-shift drive two noncommuting automorphisms of the dyadic two-solenoid $\widehat{\mathbb{Z}[1/2]^2}$, generated by $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix}$. For a chronological word w , the fibre fixed group has the intrinsic order

$$\# \text{Fix}(\alpha_{M_w}) = \text{oddp} \left(8^{|w|} - \text{tr } M_w + 1 \right).$$

The primitive words **aabbb** and **ababb** have the same letter counts but fixed-point orders 30035 and 15021, respectively. The first return has a rational zeta function, whereas the second has a natural boundary at base-clock radius $1/8$. Modulo two, the words carrying a nontrivial local correction are exactly the cyclic golden-mean language. Nevertheless, the full switching zeta does not inherit a natural boundary at its first circle: it has a meromorphic, zero-free continuation apart from one simple pole at $1/16$ throughout $|z| < (8\varphi)^{-1}$, where $\varphi = (1 + \sqrt{5})/2$. Thus orbit-resolved chronology and global aggregation exhibit different analytic behavior. The construction supplies an exact arithmetic-dynamics example but fails the periodic-clock, divisor, and operator requirements of a Hilbert–Pólya model.

1 Introduction

The area-preserving Hénon viewpoint poses a useful structural question for dynamical approaches to the Riemann zeros: after dissipation is removed, can one define an intrinsic orbit zeta whose analytic structure carries more than generic chaotic statistics? A recent project manuscript pursued this question numerically by passing from a dissipative logistic model to

$$H_a(x, y) = (1 - ax^2 - y, x), \quad \det DH_a = 1,$$

then constructing regularized quantum and Fokker–Planck solvers [11]. The manuscript carefully labels the resulting zero comparisons as numerical rather than as an identification of a Hilbert–Pólya operator. This leaves a basic mathematical gap: the conservation law alone does not provide exact periodic weights, a determinant identity, or a spectral divisor.

We address that gap by changing dynamical category. The present system is not derived from, conjugate to, or a quantization of the smooth Hénon map. It replaces nonlinear planar motion by a chronological cocycle of compact-group automorphisms. The fibre is the dyadic two-solenoid $X_2 = \widehat{\mathbb{Z}[1/2]^2}$, and a full two-shift chooses at each time between the noncommuting matrices

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix}. \tag{1.1}$$

Their real determinants are 8, so these matrices are not planar area-preserving maps. They are invertible over $\mathbb{Z}[1/2]$, however, and their dual maps are automorphisms preserving Haar measure on X_2 . This distinction is essential.

For a word $w = w_0 \cdots w_{n-1}$, later symbols multiply on the left:

$$M_w = M_{w_{n-1}} \cdots M_{w_0}. \quad (1.2)$$

We never replace this product by a power of an averaged transition matrix. Only after every ordered word has been evaluated do we form the periodic sum. This convention preserves both primitive base necklaces and the correct repetition law $M_{w^r} = M_w^r$.

The main contribution is a local–global analytic split. Chronology is strong enough to change the analytic type of two primitive return zetas with the same abelianized symbol data. Yet after all base words are summed, the full zeta crosses its leading convergence circle with no zeros. More precisely, we prove the following.

- (i) Every return is expanding and has an intrinsic finite fixed group with

$$\# \text{Fix}(\alpha_{M_w}) = \text{oddp}(\text{tr } M_w + 1).$$

The corresponding archimedean comparison has the exact rational zeta

$$Z_\infty(z) = \frac{1 - 13z + 33z^2}{(1 - 2z)(1 - 16z)}.$$

- (ii) A word receives a nontrivial dyadic correction if and only if it has no cyclic occurrence of **aa**. The active words are therefore counted by Lucas numbers. Every fixed congruence depth is represented by a finite chronological monoid recurrence.
- (iii) The primitive period-five words **aabbb** and **ababb** have the same Parikh vector $(2, 3)$, but their return zetas are respectively rational and natural-boundary. The latter boundary occurs at $|z| = 1/8$ in the base clock.
- (iv) If Δ_n is the total dyadic correction at time n , then $\lim \Delta_n^{1/n} = 8\varphi$. Consequently the full solenoid zeta continues meromorphically to $|z| < (8\varphi)^{-1}$, where its only divisor is a simple pole at $z = 1/16$.

This theorem package is not a Riemann-zero construction. Its natural clocks are the lattice scales 16, 8φ , and 8; no prime-period law, Riemann divisor, functional equation, or discrete self-adjoint operator appears. We therefore retain the arithmetic-dynamics result and reject the frozen object under the project’s Hilbert–Pólya evaluation. This negative conclusion is part of the result rather than a caveat added after the fact.

Section 2 positions the construction. Sections 3–5 establish the fixed-point formula, the parity language, and the chronology theorem. Section 6 proves the continuation disk, and Section 7 records the scoped Hilbert–Pólya ruling. Full algebraic details and reproducibility checks appear in the appendices.

2 Background and positioning

Area-preserving Hénon dynamics. Hénon’s quadratic area-preserving mappings were introduced as tractable models for two-degree-of-freedom conservative dynamics [7]. Their shear–rotation

Table 1: The source Hénon program and the present construction share a conservative motivation but are mathematically different systems. In particular, Haar preservation below is not real-plane area preservation.

Feature	Source Hénon manuscript	Present work
Phase space	$\mathbb{R}^2$	$\Sigma_2 \times \widehat{\mathbb{Z}[1/2]^2}$
Evolution	nonlinear planar map	locally constant automorphism cocycle
Conservation	$\det DH_a = 1$	Haar-preserving compact-group automorphisms
Chronology	single-map iterates; static or prescribed $a(t)$ schedules	ordered noncommuting word products
Main observable	fitted numerical spectra	intrinsic fixed groups and zeta
Operator status	heuristic solvers	no HP operator claimed

structure, invariant islands, and chaotic regions make them natural laboratories for orbit geometry. The source manuscript [11] uses a common recurrence form of this philosophy and then takes a formal continuous limit before adding a quartic confinement and numerical spectral solvers. Our use of that work is narrower: it motivates invertibility and a genuine dynamical clock. We do not transport its parameter, continuous limit, regularization, or spectral matching into the present system. This separation prevents a numerical Hénon hypothesis from becoming an unexamined premise of the exact theorems below.

Periodic points on solenoids. Dualizing modules of S -integers produces compact abelian dynamical systems whose periodic counts factor into archimedean and nonarchimedean local terms. Chothi, Everest, and Ward developed this mechanism for S -integer systems and showed that even finite S can produce irrational dynamical zeta functions [5]. Miles gave higher-dimensional periodic-point formulas for solenoids and related compact groups [9]. Lind and Ward’s entropy theory explains the corresponding decomposition into Euclidean and p -adic places [8]. The formula used here is a concrete matrix instance of that theory; because $\mathbb{Z}[1/2]$ is a localized principal ideal domain, we also prove it directly by a cokernel and Smith-normal-form argument. For rank-two solenoids specifically, Ha and Lee [6] give explicit subgroup-index and periodic-point formulas for rational matrix endomorphisms. These results place our fixed-index calculation firmly in prior theory; the new claim below concerns chronological switching and its local–global analytic split.

Dynamical zeta and natural boundaries. For a map with finite fixed sets, the Artin–Mazur series [1] packages the counts $N_n = \# \text{Fix}(F^n)$ as

$$Z_F(z) = \exp \left(\sum_{n \geq 1} \frac{N_n}{n} z^n \right). \quad (2.1)$$

Algebraic dynamical systems exhibit a sharp contrast between rational zeta functions and zeta functions whose convergence circles are natural boundaries. Bell, Miles, and Ward prove a Pólya–Carlson-type dichotomy under finite-place and archimedean hyperbolicity hypotheses [2]. We use their theorem only for a single fibre return automorphism. We do not apply it directly to the full switching skew product, and we do not infer the analytic type of an infinite product from one factor. The finite-adelically-distorted framework of Byszewski et al. [3] supplies a broader modern setting

for fixed-point sequences and zeta dichotomies, including S -integer systems. It does not by itself determine the zeta of a noncommuting full-shift cocycle.

Ordered semigroup actions. For Ruelle-expanding generators, free-semigroup dynamics, full-shift skew products, periodic-point growth, and associated zeta functions are established structures [4]. Consequently neither the use of a two-symbol driver nor the observation that noncommuting products depend on order is a novelty claim. Our result concerns a particular interaction: the mod-2 fixed-point correction selects a cyclic finite-type language; two equal-Parikh primitive returns then acquire different zeta analytic types; and the global zeta nevertheless crosses its first circle. A directed source audit did not locate this combination in one explicit solenoid cocycle, but this is a scoped assessment rather than an exhaustive priority claim.

Hilbert–Pólya scope. The broad physical motivation for a self-adjoint spectral realization of the Riemann zeros is reviewed, for example, by Schumayer and Hutchinson [10]. A conservative classical map, a unitary Koopman operator, or random-matrix-like statistics are not by themselves a Riemann divisor. Our evaluation therefore asks separately whether primitive orbits have an arithmetic clock, whether a frozen determinant has the right zeros, whether the analytic completion has the required global structure, and whether a compatible discrete self-adjoint operator exists. The present system fails those later gates despite its exact zeta theorems.

3 The chronological dyadic-solenoid system

Let $R = \mathbb{Z}[1/2]$ and let $X_2 = \widehat{R^2}$ be its Pontryagin dual. The group R^2 is torsion-free of rank two, so X_2 is a connected compact abelian group of topological dimension two. For $M \in \mathrm{GL}_2(R)$, define

$$\alpha_M = \widehat{M^\top}. \quad (3.1)$$

Pontryagin duality reverses composition; the transpose therefore gives $\alpha_M \circ \alpha_N = \alpha_{MN}$.

Let $\Sigma_2 = \{a, b\}^{\mathbb{Z}}$ with the left shift σ , and let $M_a = A$, $M_b = B$, with A, B from (1.1). We study the autonomous skew product

$$F(\omega, x) = (\sigma\omega, \alpha_{M_{\omega_0}} x). \quad (3.2)$$

Because $\det A = \det B = 8$ is a unit in R , both fibre maps are automorphisms. With the fair $(1/2, 1/2)$ Bernoulli measure on the base and Haar measure on the fibre, F is invertible and measure preserving.

If ω is fixed by σ^n , its based period block $w = w_0 \cdots w_{n-1}$ has return matrix

$$M_w = M_{w_{n-1}} \cdots M_{w_0}. \quad (3.3)$$

A cyclic rotation conjugates M_w in $\mathrm{GL}_2(R)$, and $M_{w^r} = M_w^r$. Hence fixed data descend to base necklaces while their repetitions retain the true monodromy.

Proposition 3.1 (Uniform expansion and sign). *For every word w of length n , every eigenvalue of M_w has modulus greater than one and*

$$D_w := \det(I - M_w) = 8^n - \mathrm{tr} M_w + 1 > 0. \quad (3.4)$$

If

$$\beta = \frac{7 + \sqrt{17}}{2}, \quad \gamma = \frac{7 - \sqrt{17}}{2},$$

then

$$8^n - 2\beta^n + 1 \leq D_w \leq 8^n + 1. \quad (3.5)$$

Proof. The matrices A and B are symmetric positive definite. Their least singular values are 2 and $\gamma > 1$, while their largest singular values are 4 and β . Thus $s_{\min}(M_w) \geq \gamma^n$ and $\|M_w\|_2 \leq \beta^n$. The first inequality places every eigenvalue outside the unit disk. All entries of M_w are positive. If its eigenvalues are real, their positive product and positive sum make both positive; if they are nonreal, they are a conjugate pair. In either case $\det(I - M_w) > 0$. Equation (3.4) uses $\det M_w = 8^n$, and (3.5) follows from $0 < \operatorname{tr} M_w \leq 2\|M_w\|_2$. $\square$

Theorem 3.2 (Intrinsic fixed-point index). *For every nonempty word w , the fibre fixed group is finite and*

$$\#\operatorname{Fix}(\alpha_{M_w}) = |D_w|_{\infty}|D_w|_2 = \frac{D_w}{2^{\nu_2(D_w)}} = \operatorname{oddp}(\operatorname{part}(D_w)). \quad (3.6)$$

Proof. Dualizing $I - M_w^{\top}$ identifies the fixed group with the dual of $R^2/(I - M_w^{\top})R^2$. Proposition 3.1 makes the determinant nonzero. Smith normal form over the localized principal ideal domain R shows that localization removes exactly the powers of two from the finite integer cokernel. Its remaining order is $D_w/2^{\nu_2(D_w)}$. The local-product expression is the same identity in normalized absolute-value notation. $\square$

The full fixed-point count is therefore

$$N_n^{(2)} = \#\operatorname{Fix}(F^n) = \sum_{w \in \{a,b\}^n} \operatorname{oddp}(\operatorname{part}(D_w)), \quad Z_2(z) = \exp \left(\sum_{n \geq 1} \frac{N_n^{(2)}}{n} z^n \right). \quad (3.7)$$

As an exact control, omit the 2-adic factor and count the corresponding degree-eight torus endomorphisms.

Proposition 3.3 (Archimedean comparison). *Let $N_n^{(\infty)} = \sum_{|w|=n} D_w$. Then*

$$N_n^{(\infty)} = 2^n + 16^n - \operatorname{tr}((A + B)^n) \quad (3.8)$$

and

$$Z_{\infty}(z) = \exp \left(\sum_{n \geq 1} \frac{N_n^{(\infty)}}{n} z^n \right) = \frac{1 - 13z + 33z^2}{(1 - 2z)(1 - 16z)}. \quad (3.9)$$

Proof. Distributivity gives the exact chronological identity $\sum_{|w|=n} M_w = (A + B)^n$. This is a sum of all ordered products, not a replacement of individual returns by an averaged orbit. Summing (3.4) yields (3.8). Finally, $\operatorname{tr}(A + B) = 13$, $\det(A + B) = 33$, and $\log \det(I - z(A + B)) = -\sum_{n \geq 1} \operatorname{tr}((A + B)^n) z^n / n$, which gives (3.9). $\square$

4 Symbolic parity and the congruence tower

The first local correction has an exact symbolic description. Reducing the generators modulo two gives

$$J := \bar{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad E := \bar{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad (4.1)$$

with

$$J^2 = 0, \quad E^2 = E, \quad (EJ)^2 = EJ, \quad \operatorname{tr}(EJ) = 1. \quad (4.2)$$

Theorem 4.1 (Cyclic golden-mean gate). *For every based cyclic word w ,*

$$2 \mid D_w \iff w \text{ has no cyclic occurrence of } \mathbf{aa}. \quad (4.3)$$

The number of such based words of length n is the Lucas number

$$L_n = \text{tr} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^n = \varphi^n + (-\varphi^{-1})^n. \quad (4.4)$$

Proof. Because $n \geq 1$, equation (3.4) gives $D_w \equiv 1 + \text{tr } M_w \pmod{2}$. If the cyclic word contains $\mathbf{aa}$, rotate the trace until J^2 is adjacent; its trace is then zero. If the word has no cyclic $\mathbf{aa}$, it is either all $\mathbf{b}$, whose reduction is E , or every J is separated by a nonempty E -run. Idempotence compresses the cyclic trace to $\text{tr}((EJ)^k) = 1$. This proves (4.3). Equation (4.4) is the closed-walk count for the two-state no- $\mathbf{aa}$ graph. $\square$

The higher dyadic layers remain chronological. For any positive integer d ,

$$\text{oddpair}(d) = d \left(1 - \sum_{k \geq 1} 2^{-k} \mathbf{1}_{2^k \mid d} \right). \quad (4.5)$$

Indeed, if $\nu_2(d) = m$, the finite sum in parentheses equals 2^{-m} . Define

$$C_{n,k} = \sum_{\substack{|w|=n \\ 2^k \mid D_w}} D_w. \quad (4.6)$$

Then the complete fixed count is the coefficientwise finite identity

$$N_n^{(2)} = N_n^{(\infty)} - \sum_{k \geq 1} 2^{-k} C_{n,k}. \quad (4.7)$$

Proposition 4.2 (Finite chronological recurrence at fixed depth). *For every fixed k , the sequence $C_{n,k}$ is generated by a finite linear recurrence on chronological residue states modulo 2^k .*

Proof. For a reachable residue matrix $g \in M_2(\mathbb{Z}/2^k\mathbb{Z})$, retain the word count $c_n(g)$, the scaled count $q_n(g) = 8^n c_n(g)$, and the integer matrix sum

$$S_n(g) = \sum_{\substack{|w|=n \\ M_w \equiv g \pmod{2^k}}} M_w.$$

Appending the symbol with matrix $L \in \{A, B\}$ on the right of the word multiplies the return on the left. Thus

$$c_{n+1}(h) = \sum_{Lg=h} c_n(g), \quad q_{n+1}(h) = 8 \sum_{Lg=h} q_n(g), \quad S_{n+1}(h) = \sum_{Lg=h} L S_n(g). \quad (4.8)$$

The residue $8^n \bmod 2^k$ may be included as one more finite state. The selector $2^k \mid D_w$ then depends only on that state and g , and

$$C_{n,k} = \sum_{g: 1+8^n - \text{tr } g \equiv 0 \pmod{2^k}} (c_n(g) + q_n(g) - \text{tr } S_n(g)). \quad (4.9)$$

This is a finite linear system for each fixed k . $\square$

Proposition 4.2 does not construct one finite transfer matrix for the exact odd-part weight. Equation (4.7) uses unbounded congruence depth, is exact one coefficient at a time, and supplies no license to interchange an infinite state sum with analytic continuation.

Table 2: Exact counts. The active-word column is the Lucas number from Theorem 4.1.

n	$N_n^{(\infty)}$	$N_n^{(2)}$	Δ_n	L_n
1	5	4	1	1
2	157	56	101	3
3	3194	2446	748	4
4	57121	34504	22617	7
5	969035	807804	161231	11
6	16021054	11965154	4055900	18
7	261230555	232113186	29117369	29
8	4226257633	3511221904	715035729	47
9	68064014258	63051464302	5012549956	76
10	1093258037257	972315505846	120942531411	123

5 Chronology changes analytic type

The natural orbit decomposition is by primitive base sectors, not by a dihedral quotient and not by letter incidence. If $[w]$ is a primitive base necklace of length ℓ , its invariant sector contains ℓ base phases. For $n = \ell r$, that sector contributes $\ell \# \text{Fix}(\alpha_{M_w}^r)$ to $\# \text{Fix}(F^n)$, and it contributes zero when $\ell \nmid n$. Consequently

$$Z_2(z) = \prod_{[w] \text{ primitive base}} \zeta_{\alpha_{M_w}}(z^{|w|}) \quad (5.1)$$

as an exact formal identity. Analytically, we assert the product only in its initial convergence disk. Each fibre zeta in (5.1) contains its own primitive fibre orbits; primitive base necklaces are therefore a complete base-sector decomposition, not an enumeration of all primitive F -orbits.

Let $t_w = \text{tr } M_w$, $d_w = \det M_w = 2^{3\ell}$, and write the characteristic polynomial as

$$f_w(x) = x^2 - t_w x + d_w. \quad (5.2)$$

Theorem 5.1 (Trace-parity analytic dichotomy). *Let w be a primitive base word of length ℓ .*

1. *If t_w is even, then every $\det(I - M_w^r)$ is odd and*

$$\zeta_{\alpha_{M_w}}(u) = \frac{1 - t_w u + d_w u^2}{(1 - u)(1 - d_w u)}. \quad (5.3)$$

2. *If t_w is odd, then $\zeta_{\alpha_{M_w}}(u)$ has the circle $|u| = d_w^{-1}$ as a natural boundary. The corresponding factor in the base clock has natural boundary $|z| = 1/8$.*

Proof. If t_w is even, Cayley–Hamilton gives $M_w^2 \equiv 0 \pmod{2}$, so $\det(I - M_w^r)$ is odd for every r . The solenoid fixed count equals the torus count $1 - \text{tr}(M_w^r) + d_w^r$; exponentiating its three trace terms gives (5.3).

Suppose t_w is odd. Modulo two, $f_w(x) = x(x + 1)$, with two simple roots. Hensel lifting gives one 2-adic unit eigenvalue and one eigenvalue of valuation 3ℓ , hence one isometric finite-place direction. More explicitly, the dual module is the full $\mathbb{Z}[1/2]$ -lattice $\mathbb{Z}[1/2]^2$. After passage to each characteristic-field factor, its finite place set in the Miles representation is precisely the set of places above the inverted rational prime two: localization of a full lattice adds no finite place away from two and includes every place above two. The simple factorization $f_w = x(x + 1) \pmod{2}$ and Hensel’s lemma split the scalar extension over $\mathbb{Q}_2$ into a unit line and a valuation- 3ℓ line. Equivalently, in

Table 3: Changing only chronological order changes the parity correction and the analytic type.

word	return matrix	trace	D_w	fixed count	zeta type
aabbb	$\begin{pmatrix} 1162 & 1222 \\ 1468 & 1572 \end{pmatrix}$	2734	30035	30035	rational
ababb	$\begin{pmatrix} 1133 & 1247 \\ 1422 & 1594 \end{pmatrix}$	2727	30042	15021	natural boundary

the irreducible quadratic case, two splits in the characteristic field and the unit root defines a place $v \mid 2$ in that finite place set with $|\xi|_v = 1$. In fact, the reducible case cannot occur in this odd-trace branch: rational roots of the monic integral polynomial are integers; positivity and expansion make both roots greater than one; and their product $2^{3\ell}$ then forces both roots, hence their sum t_w , to be even. Thus the finite-place condition of the natural-boundary branch is genuinely met. We now audit the remaining hypotheses of Theorem 15 of Bell et al. [2]. The group X_2 is connected and two-dimensional; d_w is a unit of $\mathbb{Z}[1/2]$, so the return is an automorphism; Proposition 3.1 excludes roots of unity and makes it ergodic; and no archimedean eigenvalue lies on the unit circle. The theorem therefore gives a natural boundary at $e^{-h(\alpha_{M_w})} = d_w^{-1}$. Here the entropy is $\log d_w$: both archimedean eigenvalues are expanding, while neither dyadic eigenvalue is 2-adically expanding. Substitution $u = z^\ell$ yields $|z| = d_w^{-1/\ell} = 1/8$. $\square$

The dichotomy becomes chronological at the first possible equal-incidence witness used here.

Theorem 5.2 (Equal-Parikh analytic-type split). *The primitive period-five words*

$$w_0 = \mathbf{aabbb}, \quad w_1 = \mathbf{ababb}$$

have the same Parikh vector $(2, 3)$ and are neither cyclic nor dihedral copies. Their return data are those in Table 3. In particular, w_0 has rational fibre zeta

$$\zeta_{\alpha_{M_{w_0}}}(z^5) = \frac{1 - 2734z^5 + 32768z^{10}}{(1 - z^5)(1 - 32768z^5)}, \quad (5.4)$$

whereas the w_1 sector has natural boundary $|z| = 1/8$.

Proof. Direct chronological multiplication gives $M_{w_0} = B^3 A^2$ and $M_{w_1} = B^2 ABA$, with the entries shown in Table 3. Both words are nonconstant of prime length and are therefore primitive. Their rotation and reversal classes are disjoint by direct inspection. Theorem 3.2 gives the fixed counts, and Theorem 5.1 gives the two analytic types. $\square$

For the natural-boundary witness, the local fluctuation is also explicit. If ξ is its 2-adic unit eigenvalue, then $\nu_2(\xi - 1) = 1$ and $\nu_2(\xi + 1) = 3$. The other eigenvalue makes $1 - \eta^r$ a unit, and the dyadic lifting-the-exponent identity gives

$$\nu_2 \det(I - M_{w_1}^r) = \begin{cases} 1, & r \text{ odd,} \\ 3 + \nu_2(r), & r \text{ even.} \end{cases} \quad (5.5)$$

The term “isometric direction” here means absolute value one at the dyadic place; it does not mean that an eigenvalue equals one.

6 The full zeta crosses its first circle

The natural-boundary sector in Theorem 5.2 does not determine the first boundary of the full product. Define the total local correction

$$\Delta_n = N_n^{(\infty)} - N_n^{(2)}. \quad (6.1)$$

By Theorem 4.1, only the L_n cyclic no-aa words contribute. For such a word, D_w is even and

$$\frac{D_w}{2} \leq D_w - \text{oddpart}(D_w) < D_w. \quad (6.2)$$

Theorem 6.1 (Exact correction scale). *Let $\varphi = (1 + \sqrt{5})/2$. Then*

$$\frac{1}{2}(8^n - 2\beta^n + 1)L_n \leq \Delta_n \leq (8^n + 1)L_n \quad (6.3)$$

for all sufficiently large n , and consequently

$$\lim_{n \rightarrow \infty} \Delta_n^{1/n} = 8\varphi. \quad (6.4)$$

Proof. Sum (6.2) over the active words and apply the uniform determinant bounds (3.5). Since $\beta < 8$ and $L_n^{1/n} \rightarrow \varphi$, taking n -th roots in (6.3) proves (6.4). $\square$

Set

$$G(z) = \sum_{n \geq 1} \frac{\Delta_n}{n} z^n, \quad R = \frac{1}{8\varphi} \approx 0.077254. \quad (6.5)$$

Theorem 6.1 makes R the exact radius of G . In the initial disk of convergence,

$$Z_2(z) = Z_\infty(z)e^{-G(z)}. \quad (6.6)$$

Theorem 6.2 (Pole-removed zero-free disk). *The function Z_2 has a unique meromorphic continuation to $|z| < R$. In this disk it has exactly one simple pole, at $z = 1/16$, and no zeros. Equivalently,*

$$H(z) := (1 - 16z)Z_2(z) = \frac{1 - 13z + 33z^2}{1 - 2z} e^{-G(z)} \quad (6.7)$$

is holomorphic and nowhere zero on $|z| < R$. The Taylor series of Z_2 at the origin has radius exactly $1/16$, but its first convergence circle is not a natural boundary.

Proof. The series G is holomorphic in $|z| < R$, so e^{-G} is holomorphic and zero-free there. Equation (6.6) therefore defines the unique meromorphic continuation from the initial disk. The numerator zeros of Z_∞ have moduli

$$\frac{2}{13 + \sqrt{37}} \approx 0.104806, \quad \frac{2}{13 - \sqrt{37}} \approx 0.289133,$$

both greater than R . Of the denominator zeros, only $1/16$ lies in the disk, and it is simple. Nothing cancels it. Formula (6.7) and all assertions follow. $\square$

Positive coefficients and the lower bound in (6.3) imply $G(r) \rightarrow +\infty$ as $r \uparrow R$. This fact does not settle the later boundary. Exponentiation can turn a logarithmic singularity of G into a zero of e^{-G} , so we do not claim a nonremovable singularity, a natural boundary, or nonrationality of Z_2 at $|z| = R$. Nor may one promote the natural boundary of a factor in (5.1) to the infinite product: cancellation and the domain of the product must first be controlled.

The local-global contrast is therefore precise. Chronology changes the analytic type of an orbit-resolved primitive base sector at the lattice scale $1/8$, but the leading full-system scale is the transparent branching-fibre pole $1/(2 \cdot 8) = 1/16$, and the dyadic correction is strictly subleading.

7 Hilbert–Pólya evaluation and conclusion

The exact arithmetic structure does not by itself make the system a Hilbert–Pólya candidate. Table 4 records the evaluation of the frozen object; it does not rule out every future reweighting or every different dynamical construction.

Table 4: Scoped Route-A evaluation. Positive evidence concerns the candidate’s intrinsic dynamics; each failure concerns the missing connection to the Riemann divisor.

Gate	Verdict	Strongest evidence	Decisive limitation
A1 orbit arithmetic	A1_WEAK	exact primitive base-sector decomposition, local weight, and repetition law	word-length clock has no prime or von Mangoldt correspondence
A2 divisor	A2_FAIL	exact zeta and proved divisor in a larger disk	no frozen $z \leftrightarrow s$ map or Riemann-zero divisor
A3 analytic structure	A3_FAIL	exact pole and continuation theorem	no functional equation, Gamma factor, or Riemann–von Mangoldt law
A4 operator	A4_FORMAL_HINT	natural unitary Koopman operator and finite congruence recurrences	no discrete self-adjoint spectrum or trace-class determinant with these weights

The operator limitation can be sharpened for the natural lift. Let $U_F f = f \circ F$ on $L^2(\Sigma_2 \times X_2, \text{Bernoulli} \times \text{Haar})$. It is unitary. The closed subspace of functions depending only on the base is invariant, and on that subspace U_F is the Koopman operator of the two-sided Bernoulli shift. If $g(\omega) = \mathbf{1}_{\{\omega_0=a\}} - 1/2$, then $\{U_F^n g : n \in \mathbb{Z}\}$ is an orthogonal family. Its cyclic subspace is a bilateral-shift component with Lebesgue spectrum. Thus the natural Koopman operator is not a pure discrete Hilbert–Pólya spectrum. Moreover, no resolvent of any bounded operator on this infinite-dimensional space can be compact: if $(U_F - \lambda I)^{-1}$ were compact, its product with $U_F - \lambda I$ would make the identity compact. This argument rejects the natural Koopman lift only; it is not a no-go theorem for every possible quantization.

The overall tuple is

$$(\text{A1_WEAK}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_FORMAL_HINT}).$$

Route B is not invoked because no compatible self-adjoint operator, determinant class, or spectral normalization has been defined.

The positive conclusion is instead a theorem in arithmetic dynamics. An explicit noncommuting solenoid cocycle has a cyclic symbolic criterion for its first local valuation, an exact congruence tower, and two equal-incidence primitive base returns with different zeta analytic types. At the same time, its full zeta continues through the leading circle and becomes zero-free once the elementary pole is removed. The example warns against two opposite shortcuts: abelianizing chronology can erase orbit-level analytic structure, whereas multiplying local natural-boundary labels can overstate the global analytic obstruction.

Two next questions are deliberately separated. Within the present system, the theorem-sized problem is to determine whether the secondary circle $|z| = (8\varphi)^{-1}$ contains a nonremovable singularity or a natural boundary. For the broader Hilbert–Pólya search, further work should change form again rather than tune this lattice pole: a nonabelian congruence voltage-graph or Ihara tower, with its trivial branching sector removed before evaluation, offers a different primitive clock and determinant architecture.

A Proof and convention details

A.1 Dual composition, rotations, and repetitions

If T is an endomorphism of a discrete abelian group, its dual acts on a character x by $\widehat{T}(x) = x \circ T$. Hence $\widehat{T} \circ \widehat{S} = \widehat{S \circ T}$. With $\alpha_M = \widehat{M^\top}$, this becomes $\alpha_M \circ \alpha_N = \alpha_{MN}$, which proves the order in (3.3).

For $w = w_0 \cdots w_{n-1}$ and its left rotation $\rho w = w_1 \cdots w_{n-1} w_0$,

$$M_{\rho w} = M_{w_0} M_w M_{w_0}^{-1}$$

in $\text{GL}_2(R)$. Thus trace, determinant, and fixed-group order are cyclic invariants. Concatenating r copies gives $M_{w^r} = M_w^r$. Because both generators are symmetric, reversing the multiplication convention transposes every word product and leaves all data used in this paper unchanged.

A.2 Localized Smith normal form

Let $C = I - M_w^\top \in M_2(\mathbb{Z})$. Over $\mathbb{Z}$, choose Smith factors $s_1, s_2 \neq 0$ with $|s_1 s_2| = |\det C| = D_w$. Tensoring the cokernel with $R = \mathbb{Z}[1/2]$ makes every power of two invertible. The finite localized cokernel therefore has order

$$\prod_{i=1}^2 \frac{|s_i|}{2^{\nu_2(s_i)}} = \frac{D_w}{2^{\nu_2(D_w)}}.$$

Pontryagin duality preserves finite cardinality and converts this cokernel into $\ker(\alpha_{M_w} - I)$, proving Theorem 3.2.

A.3 The dyadic unit direction for the period-five witness

For $M = M_{\text{ababb}}$,

$$f(x) = x^2 - 2727x + 32768.$$

The reduction $f(x) \equiv x(x+1) \pmod{2}$ has simple roots. If ξ is the unit root, then

$$f(1) = 30042, \quad \nu_2(f(1)) = 1, \quad f(-1) = 35496, \quad \nu_2(f(-1)) = 3,$$

while $f'(1)$ and $f'(-1)$ are odd. The strong Hensel estimate gives $\nu_2(\xi - 1) = 1$ and $\nu_2(\xi + 1) = 3$. The other eigenvalue has valuation 15, so $1 - \eta^r$ is a unit. For odd r , $\nu_2(\xi^r - 1) = \nu_2(\xi - 1)$. For even r , dyadic lifting-the-exponent gives

$$\nu_2(\xi^r - 1) = \nu_2(\xi - 1) + \nu_2(\xi + 1) + \nu_2(r) - 1,$$

which proves (5.5).

A.4 Why the secondary boundary is open

The lower bound (6.3) gives

$$\frac{\Delta_n}{n} R^n \geq \frac{c}{n}$$

for all sufficiently large n and some $c > 0$. Hence $G(r) \rightarrow +\infty$ along the positive real axis. It follows that the power series for G cannot be holomorphic across R . It does not follow that e^{-G} is singular there: for example, exponentiating a logarithmic singularity can yield a holomorphic zero. Proving a boundary for Z_2 would require more detailed control of the congruence layers than their exponential growth rate.

B Exact computation and reproducibility

All reported computations use Python’s exact integers and the standard library. No primes, Riemann-zero tables, ξ -function evaluations, or post-hoc spectral rescalings enter the producer. From the project directory, the frozen run is

```
python code/solenoid_zeta.py --max-period 20 \  
    --parity-period 12 --tower-level 8  
python code/independent_check.py --max-period 12  
python code/test_solenoid_zeta.py -v
```

The producer enumerates all $\sum_{n=1}^{20} 2^n = 2,097,150$ based words. It computes the ordered matrix, D_w , its dyadic valuation, and the odd-part fixed count. The period sums are checked against Proposition 3.3; the parity gate is exhausted through period 12; and the finite monoid construction is checked against direct enumeration for every $1 \leq k \leq 8$ and $1 \leq n \leq 12$.

The second program does not import the producer. It uses nested matrices and the characteristic-trace recurrence to recompute all counts through period 12 and the witness repetitions through $r = 20$. Ten deterministic unit tests cover the frozen matrices, product order, index formula, mod-2 language, period-five analytic split, repetition valuation, archimedean closed form, Dold integrality, integral zeta coefficients, and a rational-collapse control.

If P_n denotes the number of primitive F -orbits of total period n , the Dold inversion

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) N_{n/d}^{(2)}$$

is a nonnegative integer for every $n \leq 20$. This ledger checks the total periodic counts; it is distinct from the primitive base-sector decomposition in (5.1).

The result directory contains the frozen certificate, periodic counts, valuation distributions, congruence layers, witness matrices, the independent report, and SHA-256 hashes. A Berlekamp–Massey screen of the first 20 coefficients is also persisted, but no theorem uses that finite-prefix experiment.

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